

State of the Art and Technical References

Agricultural social navigation, reactive families, and implementation scope

A synthesis aligned with the corrected manuscript's thirteen references. It distinguishes canonical DWA, ORCA/RVO, Social Force, and CBF principles from the evaluated B1-B4 implementations and corrects the interpretation of the M4 field.

Study	9 methods × 4 scenarios × 10 paired repetitions
Navigation data	360 valid runs; 360 successful; 0 excluded; 0 failed
Source	Corrected manuscript supplied for this update (13 September 2026)

Agri-Social Navigation in Vineyard Corridors

1. Scope

Social mobile-robot navigation combines physical avoidance, separation regulation, legibility, efficiency, and operational continuity. In vineyards, narrow geometry limits lateral deviations and increases the importance of human perception and local response. References [2,6,8] organize the field around rules, costs, social fields, multi-agent interaction, optimization, and learning.

2. Proxemics and orientation

Hall's proxemics [1] distinguishes intimate, personal, social, and public regions. Robotics uses these as operational references rather than universal boundaries. The study adopts 0.45 m, 1.20 m, and 3.60 m. The literature recognizes that orientation, context, culture, and relative motion can change the shape of human space [2,9].

Zone	Range	Use in this study
Intimate	$d < 0.45$ m	Critical close-range threshold
Personal	$0.45 \leq d < 1.20$ m	Primary exposure interval
Social	$1.20 \leq d < 3.60$ m	Anticipatory interaction range
Public	$d \geq 3.60$ m	Nominal navigation region

3. Reactive navigation families

Family	Canonical principle	Evaluated role	Claim boundary
DWA / B1	Reachable command sampling and scoring [5]	Social DWA with proxemic costs	Serialized study configuration; not universal defaults
ORCA/RVO / B2	Reciprocal velocity-space collision avoidance [4]	Velocity-obstacle-risk correction	ORCA/RVO-labelled implementation; not exact canonical reproduction
Social Force / B3	Goal and interaction forces [3]	Continuous goal plus repulsion	Parameters characterize this implementation
CBF / B4	Safe-set constraints over nominal control [10]	Sampled candidate plus discrete barrier filter	CBF-Social-DWA study variant
Proxemic field / M4	Orientation-dependent spatial influence [9]	Anisotropic nearest-human field and speed regulation	Front-rear symmetric; no escape-speed guarantee

4. Final M4 formulation

$$C_{\text{ani}}(d, \Delta) = \exp\{-d^2[\cos^2(\Delta)/\sigma_L^2 + \sin^2(\Delta)/\sigma_T^2]\}$$

Final-source values are $\sigma_L = 1.20$ m, $\sigma_T = 0.80$ m, avoidance gain 3.0, and influence distance 4.0 m. The field is applied to the nearest detected human. Nominal speed is 1.0 m/s outside the social zone, approximately 0.6075 m/s in the social zone, 0.45 m/s in the personal zone, and 0 in the intimate zone when intimate stopping is enabled.

$$\cos^2(\Delta + \pi) = \cos^2(\Delta), \sin^2(\Delta + \pi) = \sin^2(\Delta)$$

The equation distinguishes longitudinal and lateral directions but gives the same value in front and behind. M4 is therefore longitudinal-lateral anisotropic with front-rear symmetry. The final code does not establish a minimum escape-speed safeguard or multi-human influence aggregation.

5. B1-B4 selection and fairness

The four families were selected to cover complementary mechanisms: social-cost command sampling, relative-velocity risk, virtual forces, and barrier filtering. Fairness is based on a common mission, environment, scenario order, seeds, timeout, telemetry, and metrics, with method-specific parameters fixed before the 360 outcomes were inspected.

Method	Parameter	Final value	Traceability
B1	Speed / angular limit	0-1.0 m/s / 1.35 rad/s	Serialized SocialDWAConfig

Method	Parameter	Final value	Traceability
B1	Samples / horizon / step	10 × 21 / 2.5 s / 0.10 s	210 commands; 25 rollout steps
B1	Cost weights	1.0/0.3/0.5/2.0/0/3.0	Implementation settings
B2	Neighbor distance / horizon	5.0 m / 3.0 s	Final source settings
B2	Avoidance / emergency gain	1.25 / 2.5	Final source settings
B3	Relaxation / force / range	0.60 s / 2.25 / 0.85 m	Final source settings
B4	Horizon / step / samples	2.0 s / 0.10 s / 8 × 17	136 commands; 20 steps
B4	Safe distance / barrier gain	1.20 m / 1.50	Final source settings

References justify algorithmic families and structures; they do not show that each numerical value is universally recommended by the original papers. Equal historical tuning effort is not claimed.

6. Evaluation and sim-to-real scope

A social-navigation comparison requires safety, exposure, efficiency, and kinematic behavior together with scenario effects [6,8,11,12]. The campaign uses run-level metrics and Method × Scenario analysis. The separate 54-run computational benchmark characterizes kernel latency, effective invocation frequency, and Unity Editor process CPU; it is not a real-time guarantee for a deployed robot.

Table 3. Dedicated benchmark. Latency columns are in ms; Hz is effective invocation frequency and CPU is whole-process Unity Editor utilization.

Method	Mean	Median	P95	P99	Max	Hz	CPU
M0	0.0216	0.0185	0.0292	0.0850	0.7095	5.47	21.17
M1	0.0215	0.0186	0.0297	0.0948	0.4413	5.46	21.28
M2	0.0220	0.0192	0.0297	0.0762	1.0879	5.46	21.35
M3	0.0224	0.0190	0.0310	0.0919	2.4147	5.46	21.20
M4	0.0213	0.0184	0.0291	0.0755	1.7663	5.49	21.26
B1	10.6138	10.2651	14.1029	17.3507	37.8071	50.03*	22.28
B2	0.0548	0.0484	0.0813	0.1526	2.3469	5.50	21.31
B3	0.0545	0.0487	0.0777	0.1621	1.8658	5.46	21.24
B4	2.7531	2.4944	4.8072	6.8417	12.1060	5.49	21.38

* B1 executes Social DWA in FixedUpdate; 50.03 Hz is the aggregate effective rate. In B1, 136 of 29,522 invocations (0.461%) exceeded 20 ms. No hard real-time guarantee is inferred.

- LiDAR/RGB-D noise and human-orientation estimation
- odometry error, actuation delay, and braking dynamics
- wheel slip and uneven terrain
- communication latency and vegetation occlusion
- adaptive human behavior not fully represented

7. Manuscript references

- [1] Hall, E. T. The Hidden Dimension. Doubleday, 1966.
- [2] Kruse, T.; Pandey, A. K.; Alami, R.; Kirsch, A. Human-aware robot navigation: A survey. *Robotics and Autonomous Systems* 2013, 61, 1726-1743. doi:10.1016/j.robot.2013.05.007.
- [3] Helbing, D.; Molnar, P. Social force model for pedestrian dynamics. *Physical Review E* 1995, 51, 4282-4286. doi:10.1103/PhysRevE.51.4282.
- [4] van den Berg, J.; Guy, S. J.; Lin, M.; Manocha, D. Reciprocal n-body collision avoidance. *Robotics Research* 2011, 70, 3-19. doi:10.1007/978-3-642-19457-3_1.
- [5] Fox, D.; Burgard, W.; Thrun, S. The dynamic window approach to collision avoidance. *IEEE Robotics and Automation Magazine* 1997, 4, 23-33. doi:10.1109/100.580977.
- [6] Singamaneni, P. T. et al. A survey on socially aware robot navigation: Taxonomy and future challenges. *International Journal of Robotics Research* 2024, 43, 1533-1572. doi:10.1177/02783649241230562.

- [7] Brito, B.; Floor, B.; Ferranti, L.; Alonso-Mora, J. Model Predictive Contouring Control for Collision Avoidance in Unstructured Dynamic Environments. *IEEE Robotics and Automation Letters* 2019, 4, 4459-4466. doi:10.1109/LRA.2019.2929976.
- [8] Mavrogiannis, C. et al. Core Challenges of Social Robot Navigation: A Survey. *ACM Transactions on Human-Robot Interaction* 2023, 12, 1-39. doi:10.1145/3583741.
- [9] Truong, X. T.; Ngo, T. D. Toward Socially Aware Robot Navigation in Dynamic and Crowded Environments: A Proactive Social Motion Model. *IEEE Transactions on Automation Science and Engineering* 2017, 14, 1743-1760. doi:10.1109/TASE.2017.2731371.
- [10] Ames, A. D. et al. Control barrier functions: Theory and applications. *European Control Conference* 2019, 3420-3431. doi:10.23919/ECC.2019.8796030.
- [11] Kang, S. et al. Social Type-Aware Navigation Framework for Mobile Robots in Human-Shared Environments. *Sensors* 2024, 24, 4862. doi:10.3390/s24154862.
- [12] Scales, P.; Aycard, O.; Aubergé, V. Planning Socially Expressive Mobile Robot Trajectories. *Sensors* 2024, 24, 3533. doi:10.3390/s24113533.
- [13] Betancourt, R. et al. Agri-Social-Nav-Vineyard: Anisotropic Proxemic Fields for Social Navigation in Agricultural Vineyard Corridors. *GitHub repository*, 2026. <https://github.com/rjbetancourt/agri-social-nav-vineyard>.